TUTORIAL 2

Proof Methods, Sets, and Fields

MOTIVATION AND GUIDING IDEA

A proof must work within the given setting: equality of sets is proved by two inclusions, and a field identity is proved from the field axioms. In a finite field, addition, subtraction, multiplication, and division by nonzero elements are permitted; numerical notation does not imply an ordering like that of $\mathbb{R}$.

NOTATION AND CONVENTIONS

The symbol $x \in A$ expresses membership of an element, whereas $A \subseteq B$ expresses inclusion of sets. The quantifiers $\forall$ and $\exists$ mean ``for every'' and ``there exists''; their order matters. The set $A \setminus B$ consists of elements of A absent from B, and $|A|$ is its cardinality when A is finite. For a field F, write $F^\times = F \setminus \{0\}$. The notation $a \equiv b \pmod{p}$ means $p \mid (a-b)$. For prime p, writing $\mathbb{Z}_p = \{0, \ldots, p-1\}$ uses representatives of residue classes. The symbol $\mathbb{F}_q$ denotes a field with q elements; when $q = p^n$ with $n > 1$, it must not be identified with arithmetic modulo q.

DEFINITION 2.1

Equality of sets and cancellation. For sets A, B, we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. To prove equality, take an arbitrary element from each side and prove that it belongs to the other side. In a field F, for $a, b, c \in F$ with $c \neq 0$,

$$ac = bc \quad \Longleftrightarrow \quad a = b,$$

because multiplication by c^{-1} is permitted. The condition $c \neq 0$ is essential. The characteristic of a field is the least positive integer p for which $p \cdot 1 = 0$, if such an integer exists; otherwise its characteristic is 0. The characteristic of a finite field is prime.

MOTIVATION AND GUIDING IDEA

Objectives. We will strengthen proofs by induction and proofs of equivalence, identify the field axioms, and calculate in finite fields. We will distinguish familiar real arithmetic from arithmetic adapted to the specified field.

The first part is a focused review of proof methods and the language of sets. The second part introduces fields and finite field arithmetic; the order used in class will follow the progress of the lectures.

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2.1 The induction principle

DEFINITION 2.2

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 2.1

Please prove by induction that $2 + 4 + \cdots + 2n = n(n+1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $2 = 1 \cdot 2$. Suppose $\sum_{j=1}^{k} 2j = k(k+1)$ for $k \geq 1$. Then

$$\sum_{j=1}^{k+1} 2j = k(k+1) + 2(k+1) = (k+1)(k+2).$$

Induction proves the identity for every $n \geq 1$.

EXERCISE 2.2

Please prove by induction that $17 \mid (18^n - 1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $18 - 1 = 17$. Suppose $18^k - 1 = 17m$ for some $m \in \mathbb{Z}$. Then

$$18^{k+1} - 1 = 18(18^k - 1) + 17 = 17(18m + 1).$$

This is divisible by 17, completing the induction step and the proof.

EXERCISE 2.3

Please prove by induction that $4^n \geq 3n^2$ for every $n \geq 3$.

WORKED SOLUTION

For $n = 3$, $4^3 = 64 \geq 27 = 3 \cdot 3^2$. Suppose $4^k \geq 3k^2$ for $k \geq 3$. Then

$$4^{k+1} \geq 12k^2.$$

Since $4k^2 - (k+1)^2 = 3k^2 - 2k - 1 = (3k+1)(k-1) > 0$,

$$4^{k+1} \geq 12k^2 \geq 3(k+1)^2.$$

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EXERCISE 2.3 CONTINUED

The induction proves the claim for every $n \geq 3$.

2.2 Sets

DEFINITION 2.3

- **Inclusion:** $A \subseteq B$ means that every element of A also belongs to B.
- **Union:** $A \cup B = \{\, x \mid x \in A \text{ or } x \in B \,\}$.
- **Intersection:** $A \cap B = \{\, x \mid x \in A \text{ and } x \in B \,\}$.

$$\Omega = \{2, 4, 6, 8, 10\}, \quad A = \{2, 4, 6\}, \quad B = \{4, 6, 8\}$$

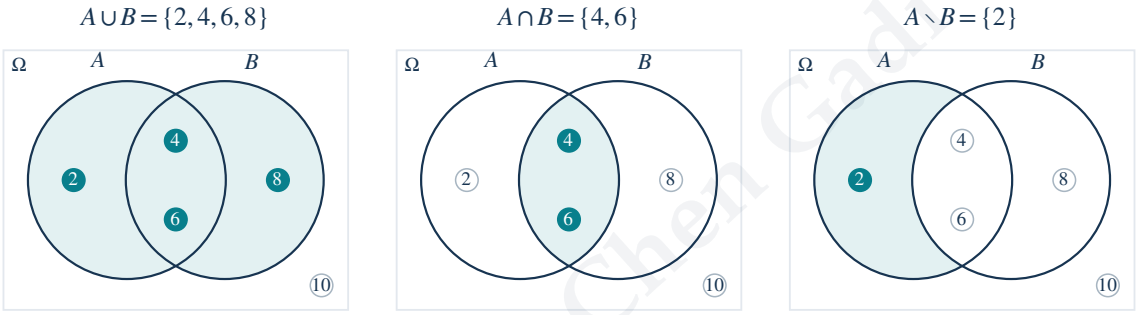


Figure 2.1. For $A = \{2, 4, 6\}$ and $B = \{4, 6, 8\}$ in $\Omega = \{2, 4, 6, 8, 10\}$, the shaded regions show $A \cup B$, $A \cap B$, and $A \setminus B$, respectively. Each numeral's position records its membership; 10 belongs to the universal set but to neither A nor B.

EXERCISE 2.4

Please prove that $A \subseteq B$ if and only if $A \cup B = B$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. The definition of union gives $A \cup B \subseteq B$, while $B \subseteq A \cup B$ always holds. Therefore $A \cup B = B$.
Second direction: suppose $A \cup B = B$. Let $x \in A$. Then $x \in A \cup B$, so $x \in B$. Therefore $A \subseteq B$.

EXERCISE 2.5

Please prove that $A \subseteq B$ if and only if $A \cap B = A$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. If $x \in A \cap B$, then $x \in A$ and $x \in B$, so $A \cap B \subseteq A$. Conversely, if $x \in A$, then the inclusion gives $x \in B$. Thus $x \in A \cap B$, proving $A \subseteq A \cap B$. Hence $A \cap B = A$.

Second direction: suppose $A \cap B = A$. Let $x \in A$. Then $x \in A \cap B$, so $x \in B$. Therefore $A \subseteq B$.

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2.3 Algebraic structure: fields

2.3.1 The field axioms

RECALL AND TOOLS

A field is a set F equipped with two operations, addition and multiplication. For all $a, b, c \in F$, the operations satisfy the following axioms. In particular, $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$ is required:

- **Closure:** if $a, b \in F$, then $a +_{\mathbb{F}} b \in F$ and $a \times_{\mathbb{F}} b \in F$.
- **Commutativity:** $a +_{\mathbb{F}} b = b +_{\mathbb{F}} a$ and $a \times_{\mathbb{F}} b = b \times_{\mathbb{F}} a$.
- **Associativity:** $(a +_{\mathbb{F}} b) +_{\mathbb{F}} c = a +_{\mathbb{F}} (b +_{\mathbb{F}} c)$ and $(a \times_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} (b \times_{\mathbb{F}} c)$.
- **Distributivity:** $a \times_{\mathbb{F}} (b +_{\mathbb{F}} c) = a \times_{\mathbb{F}} b +_{\mathbb{F}} a \times_{\mathbb{F}} c$ and $(a +_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} c +_{\mathbb{F}} b \times_{\mathbb{F}} c$.
- **Identity elements:** there is $0_{\mathbb{F}} \in F$ such that $a +_{\mathbb{F}} 0_{\mathbb{F}} = a$ for every $a \in F$, and there is $1_{\mathbb{F}} \in F$ such that $a \times_{\mathbb{F}} 1_{\mathbb{F}} = a$ for every $a \in F$.
- **Additive and multiplicative inverses:** every $a \in F$ has an additive inverse $-a$ with $a +_{\mathbb{F}} (-a) = 0_{\mathbb{F}}$; every nonzero $a \in F$ has a multiplicative inverse a^{-1} with $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXAMPLE 2.1

- **Examples:** $\mathbb{Q}$ (rational numbers), $\mathbb{R}$ (real numbers), and $\mathbb{C}$ (complex numbers) are fields with their usual addition and multiplication.
- **Nonexamples:** $\mathbb{Z}$ and $\mathbb{N}$ are not fields, because most of their elements have no multiplicative inverse; for example, 3 has no inverse in $\mathbb{Z}$.

2.3.2 Notation

A general field is usually denoted by F (sometimes $\mathbb{F}$). When no particular field is intended, we write ``Let F be a field.''
Notation convention: when the field is clear from context, we omit its subscript from the operations and the elements $0, 1$.

EXERCISE 2.6

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that $(a^{-1})^{-1} = a$.

WORKED SOLUTION

Recall that every invertible field element has a unique multiplicative inverse, and in particular $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$. Put $x = (a^{-1})^{-1}$ and calculate:

$$(a^{-1}) \times_{\mathbb{F}} x = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a$$

$$a \times_{\mathbb{F}} (a^{-1}) \times_{\mathbb{F}} x = a \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} x = a \qquad (*)$$

Thus $x = a$, that is, $(a^{-1})^{-1} = a$.
Remark: step $(*)$ uses $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

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EXERCISE 2.7

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that the multiplicative inverse of a is unique.

WORKED SOLUTION

Suppose $b \in F$ satisfies $a \times_{\mathbb{F}} b = 1_{\mathbb{F}}$. Multiply on the left by a^{-1}:

$$a \times_{\mathbb{F}} b = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a^{-1}$$

$$(a^{-1}) \times_{\mathbb{F}} a \times_{\mathbb{F}} b = a^{-1} \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} b = a^{-1} \qquad (*)$$

Hence $b = a^{-1}$, so the inverse of a is unique. *Remark:* again, step $(*)$ uses $(a^{-1}) \times_{\mathbb{F}} a = 1_{\mathbb{F}}$.

2.4 Further exercises on fields

EXERCISE 2.8

Let a, b be elements of a field $\mathbb{F}$. Please prove that if $a \times_{\mathbb{F}} b = 0_{\mathbb{F}}$, then $a = 0_{\mathbb{F}}$ or $b = 0_{\mathbb{F}}$.

WORKED SOLUTION

We distinguish two cases according to b. **Case 1:** $b = 0_{\mathbb{F}}$, and the conclusion already holds. **Case 2:** $b \neq 0_{\mathbb{F}}$. Since $\mathbb{F}$ is a field, b has an inverse b^{-1}. Multiplying both sides by b^{-1} gives

$$a \times_{\mathbb{F}} b \times_{\mathbb{F}} b^{-1} = 0_{\mathbb{F}} \times_{\mathbb{F}} b^{-1} \Longrightarrow a \times_{\mathbb{F}} 1_{\mathbb{F}} = a = 0_{\mathbb{F}}$$

PROPOSITION 2.1

A field has no nontrivial zero divisors.

EXERCISE 2.9

Let F be a field. On F^3, define

$$(a, b, c) \oplus (d, e, f) = (a + d, b + e, c + f),$$

$$(a, b, c) \otimes (d, e, f) = (ad, be, cf).$$

Please determine whether this is a field.

WORKED SOLUTION

The zero element is $(0, 0, 0)$, but

$$(1, 0, 0) \otimes (0, 1, 0) = (0, 0, 0).$$

Neither factor is zero, since $1 \neq 0$ in F. A field has no nontrivial zero divisors, so this structure is not a field.

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2.5 Finite fields

NOTE

For a prime p, we write $\mathbb{Z}_p = \{0, \dots, p-1\}$ and perform addition and multiplication modulo p. These integers represent residue classes. The same field is also denoted $\mathbb{F}_p$. Most finite-field examples here use these residue fields.

Definition and notation

DEFINITION 2.4

- A **finite field** is a field with finitely many elements. Write $|\mathbb{F}| = q$.
- Every finite field has $q = p^n$ elements, where p is prime and $n \geq 1$ is an integer. This is a structural theorem about finite fields.
- For a prime p, the set $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$, with addition and multiplication modulo p, is a field. It is called the **prime field of characteristic** p.
- Addition and multiplication in $\mathbb{Z}_p$ are defined by

$$a +_{\mathbb{Z}_p} b := (a + b) \bmod p, \quad a \times_{\mathbb{Z}_p} b := (a \cdot b) \bmod p.$$

Examples

EXAMPLE 2.2

- $\mathbb{Z}_2 = \{0, 1\}$, with $1 + 1 = 0$ and $1 \cdot 1 = 1$.
- $\mathbb{Z}_7 = \{0, 1, \dots, 6\}$, with arithmetic modulo 7; for example, $5 + 4 = 2$ and $3 \cdot 5 = 1$ in the field.
- $\mathbb{Z}_{11} = \{0, 1, \dots, 10\}$, with arithmetic modulo 11; for example, $8 + 7 = 4$ and $4 \cdot 3 = 1$ in the field.

Basic properties

PROPOSITION 2.2

In every finite field $\mathbb{F}_q$, the set $\mathbb{F}_q^\times$ of nonzero elements is closed under multiplication, and each of its elements has a unique inverse.

PROPOSITION 2.3

In every finite field $\mathbb{F}_q$, each $a \in \mathbb{F}_q$ satisfies

$$a^q = a.$$

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PROPOSITION 2.4

Every nonzero a in a finite field $\mathbb{F}_q$ satisfies $a^{q-1} = 1$.

Explanation: Closure follows because a field has no zero divisors. The existence and uniqueness of inverses follow from the field axioms and the proof above. To prove both power identities, let $a \neq 0$ and put

$$P = \prod_{x \in \mathbb{F}_q^\times} x \neq 0.$$

Multiplication by a permutes the nonzero elements: the map $x \mapsto ax$ is invertible, with inverse $x \mapsto a^{-1}x$. Thus

$$P = \prod_{x \in \mathbb{F}_q^\times} (ax) = a^{q-1}P.$$

Since $P \neq 0$, cancellation is permitted and gives $a^{q-1} = 1$, hence $a^q = a$. For $a = 0$, the identity $0^q = 0$ is immediate.

EXAMPLE 2.3

The nonzero inverses in $\mathbb{Z}_{11}$ are

a	1	2	3	4	5	6	7	8	9	10
a^{-1}	1	6	4	3	9	2	8	7	5	10

For example, $2 \cdot 6 = 12 \equiv 1$, $5 \cdot 9 = 45 \equiv 1$ and $7 \cdot 8 = 56 \equiv 1 \pmod{11}$.

EXERCISE 2.10

Please show that, in every finite field $\mathbb{F}_q$,

$$(a - b)^q = a^q - b^q.$$

WORKED SOLUTION

Use the identity $x^q = x$, proved for every $x \in \mathbb{F}_q$. In particular, it applies to $a - b$, so

$$(a - b)^q = a - b = a^q - b^q.$$

This proves the claim directly, without calculating binomial coefficients.

EXERCISE 2.11

Please work in $\mathbb{Z}_{17}$.

1. Please find the inverses of $2, 3, 5, 7, 8$.
2. Please calculate $E = (7^{-1} + 3)(5 - 2^{-1})^{-1} - 2$.

WORKED SOLUTION

All arithmetic is modulo 17. We have

$$2 \cdot 9 = 18 \equiv 1, \quad 3 \cdot 6 = 18 \equiv 1, \quad 5 \cdot 7 = 35 \equiv 1,$$

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EXERCISE 2.11 CONTINUED

$$7 \cdot 5 = 35 \equiv 1, \qquad 8 \cdot 15 = 120 \equiv 1.$$

Thus the inverses, in order, are $9, 6, 7, 5, 15$. Now $7^{-1} + 3 = 5 + 3 = 8$, while

$$5 - 2^{-1} = 5 - 9 = -4 \equiv 13 \neq 0.$$

The inverse of 13 is 4, because $13 \cdot 4 = 52 \equiv 1$. Therefore

$$E = 8 \cdot 4 - 2 = 30 \equiv 13 \pmod{17}.$$

EXERCISE 2.12

Please show that for every nonzero a in a finite field $\mathbb{F}_q$, there is $n \in \mathbb{N}$ with $a^n = 1$.

WORKED SOLUTION

The set $\mathbb{F}_q^\times$ contains exactly $q - 1$ elements and is closed under multiplication. Put $m = q - 1$ and consider the list

$$a, a^2, a^3, \ldots, a^{m+1}.$$

It has $m + 1$ entries but only m possible distinct values, so two entries are equal, say $a^k = a^l$ with $k > l$. Multiplying by a^{-l} gives $a^{k-l} = 1$. Thus every nonzero element has a positive integer power equal to the multiplicative identity.

2.6 Calculations and applications in finite fields

EXERCISE 2.13

Let $a, b, c, d \in \mathbb{Z}_2$ be four bits. Add a bit e such that $a + b + c + d + e = 0$. Please find e and please calculate it for $(a, b, c, d) = (1, 0, 1, 1)$. Please determine whether the check detects one flipped bit and what happens with two flipped bits.

WORKED SOLUTION

In $\mathbb{Z}_2$, $-x = x$, so $e = a + b + c + d$. For the given example, $e = 1 + 0 + 1 + 1 = 1$. Write the five error bits as $\varepsilon_1, \ldots, \varepsilon_5$, with 1 indicating a flip and 0 no flip. The received parity is

$$(a + b + c + d + e) + \sum_{j=1}^{5} \varepsilon_j = \sum_{j=1}^{5} \varepsilon_j.$$

A single flip gives parity 1 and is detected. Two flips give $1 + 1 = 0$ and are not detected. The check does not identify which bit flipped.

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Additional practice

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EXERCISE 2.14 FOUNDATION

In $\mathbb{Z}_{11}$, define

$$A = \{x \in \mathbb{Z}_{11} : 3x + 5 = 2\}, \qquad B = \{x \in \mathbb{Z}_{11} : x^2 = 1\}.$$

Please find $A, B, A \cup B, A \cap B$ and please determine whether $A \subseteq B$.

WORKED SOLUTION

All arithmetic is modulo 11. The first equation gives $3x = 8$. Since $3^{-1} = 4$, we get $x = 4 \cdot 8 = 32 = 10$, so $A = \{10\}$. The second equation is equivalent to $(x-1)(x+1) = 0$. A field has no zero divisors, so $x = 1$ or $x = -1 = 10$. Hence

$$B = \{1, 10\}, \quad A \subseteq B, \quad A \cup B = \{1, 10\}, \quad A \cap B = \{10\}.$$

EXERCISE 2.15 CHALLENGE

Let F be a finite field with $q > 2$ elements. Please prove that

$$\sum_{x \in F} x = 0, \qquad \sum_{x \in F^\times} x^{-1} = 0.$$

Please check both statements in the field with two elements. Do not assume an ordering on F.

WORKED SOLUTION

Put $S = \sum_{x \in F} x$. Since $q > 2$, choose $c \in F \setminus \{0, 1\}$. The map $x \mapsto cx$ is bijective, with inverse $x \mapsto c^{-1}x$. It therefore only rearranges the summands, and

$$cS = \sum_{x \in F} cx = \sum_{x \in F} x = S.$$

Thus $(c-1)S = 0$. Since $c - 1 \neq 0$, field cancellation gives $S = 0$. Removing the summand 0 does not change the sum, so $\sum_{x \in F^\times} x = 0$. The map $x \mapsto x^{-1}$ also permutes $F^\times$, because applying it twice returns x. Therefore

$$\sum_{x \in F^\times} x^{-1} = \sum_{x \in F^\times} x = 0.$$

In $\mathbb{Z}_2$, we have $0 + 1 = 1 \neq 0$, and the sum of the nonzero inverses is $1^{-1} = 1 \neq 0$. Indeed, no element c distinct from $0, 1$ exists there, so the choice used in the proof is unavailable.

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EXERCISE 2.16 CHALLENGE

Let F be a field satisfying $x^3 = x$ for every $x \in F$. Please prove that F has either two or three elements. Please write its operations in each case and please identify them with those of $\mathbb{Z}_2$ or $\mathbb{Z}_3$.

WORKED SOLUTION

For $x \neq 0$, cancellation gives $x^2 = 1$, hence $(x-1)(x+1) = 0$. A field has no zero divisors, so $x = 1$ or $x = -1$. Thus $F \subseteq \{0, 1, -1\}$. If $1 = -1$, then $F = \{0, 1\}$ and $1 + 1 = 0$. Together with multiplication by zero and $1 \cdot 1 = 1$, these are the operations of $\mathbb{Z}_2$. If $1 \neq -1$, all three elements are distinct and present. The sum $1 + 1$ cannot be 0, which would give $1 = -1$, or 1, which would give $1 = 0$ by cancellation. Therefore $1 + 1 = -1$. Writing $2 = -1$, we obtain

$$1 + 2 = 0, \qquad 2 + 2 = 1, \qquad 2 \cdot 2 = 1.$$

These, together with the zero and identity rules, give exactly $\mathbb{Z}_3$. Conversely, direct substitution of $0, 1$ in $\mathbb{Z}_2$, and $0, 1, -1$ in $\mathbb{Z}_3$, verifies $x^3 = x$.

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EXERCISE 2.17 CHALLENGE

Let F be a finite field. Please prove

$$\prod_{x \in F \setminus \{0\}} x = -1.$$

Here $\prod$ multiplies all nonzero elements. Please also prove that, for every $b \in F$,

$$\prod_{a \in F \setminus \{b\}} (b - a) = -1.$$

WORKED SOLUTION

Pair each nonzero x with x^{-1}, except when the two are equal. Each pair contributes 1 to the product. The unpaired elements satisfy $x = x^{-1}$, equivalently

$$x^2 = 1, \qquad (x-1)(x+1) = 0.$$

Since a field has no zero divisors, these are exactly 1 and -1. If they are distinct, their product is -1. If $1 = -1$, only the element 1 remains unpaired, and its value is again -1. This proves the first identity in every characteristic. For fixed b, the values $b - a$ as a ranges over $F \setminus \{b\}$ run through every nonzero field element exactly once. The second product is therefore the same product in a different order.

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